Duy Nguyen Cao

Ho Chi Minh City | 0908 814 412 | caoduynguyen12@gmail.com | linkedin.com/in/duy-nguyen-2b4883336 | github.com/Pongaaa

PROFESSIONAL SUMMARY

3rd-year Computer Science student at UIT and Top 6 VLSP 2025 Finalist. Experienced in building end-to-end Machine Learning solutions, with a focus on Recommendation Systems and Speech Processing pipelines.

EDUCATION & CERTIFICATES

University of Information Technology National University, HCM city

Bachelor of Computer Science *Expected December 2026*

- Applications of AI for Anomaly Detection – NVIDIA (November 2025)
- TOEIC 795/990 (July 2024)
- GPA: 8.36/10

PROJECT

Product Recommendation System January 2024 – February 2025

Technologies: Python, Polars, LightGBM, Collaborative Filtering *Academic Project*

- Engineered customer behavior features and cleaned real-world transaction data from the Concung dataset using Polars, optimizing the data pipeline for large-scale processing.
- Developed a Top-10 product recommendation system by combining Collaborative Filtering with classification models, achieving a 14% Precision@10 on ground-truth data.
- Conducted in-depth Exploratory Data Analysis (EDA) to identify purchasing trends, providing actionable insights that supported inventory optimization and personalized marketing strategies.

GitHub: github.com/Pongaaa/Top10-Item-Recommendation-System

Subscription Prediction System September 2025 – Present

Technologies: Python, Flask, Power BI, Logistic Regression, Random Forest *Academic Project*

- Analyzed the Shopping Behavior Dataset to identify key predictors of user intent, performing in-depth EDA to address data imbalance and improve model robustness.
- Trained and optimized classification models (Random Forest, Logistic Regression) to predict user subscription likelihood, reducing ineffective sales outreach time by 30–40%.
- Built interactive Power BI dashboards to visualize customer segments and behavior patterns, enabling decisions based on data for the sales team.
- Deployed a real-time prediction service via a Flask API integrated with a web interface, allowing the sales team to monitor and interact with model results seamlessly.

GitHub: github.com/Pongaaa/Predict-User-Subscription

COMMUNITY & LEADERSHIP

VLSP 2025 Competition June 2025 – August 2025

Research Member *Online*

- Ranked Top 6 in the prestigious VLSP 2025 national competition, achieving a competitive 75.34% accuracy on the private test set for both transcription and emotion classification.
- Architected a hybrid Automatic Speech Recognition (ASR) & Speech Emotion Recognition (SER) system for the Vietnamese language, effectively addressing complex audio processing challenges in a high-pressure environment.
- Streamlined end-to-end preprocessing and model inference pipelines on Google Colab, significantly enhancing execution speed and efficiency for large-scale audio datasets.

TECHNICAL SKILLS

Programming: Python (Polars, Pandas, NumPy, Scikit-learn, LightGBM, PyTorch, Matplotlib, Seaborn), SQL, C++, JavaScript.
Tools & Databases: SQL Server, MySQL, MongoDB, Git, GitHub, Power BI, VS Code, Kaggle, Google Colab.
Specialties: EDA, Data Visualization, Machine Learning (Regression, Random Forest, Collaborative Filtering), Recommendation Systems, Vector Search, Feature Engineering.